EXPLORATORY DATA ANALYSIS ON A CARS DATASET

Used Car Dataset

Submitted by

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Batch Number: ____________________

Project detail	Value
Dataset	Used Car Dataset
Source	Kaggle
Original records	9,582
Final cleaned records	8,812
Columns	11
Analysis	Exploratory Data Analysis
Machine learning	Not required

Clean, explore, visualize, and explain meaningful patterns in automobile data.

1. Objective

The objective is to clean, explore, visualize, and explain meaningful patterns in used-car listing data. The analysis focuses on vehicle age, model year, kilometres driven, asking price, brand, transmission, fuel type, and ownership.

Questions explored

- How are vehicle years, ages, mileage, and asking prices distributed?
- Which brands contribute the most listings?
- How does asking price vary across brands and transmission types?
- What relationships exist between age, model year, mileage, and asking price?

2. Dataset Overview

The public Indian used-car dataset contains 9,582 listing records and 11 columns: Brand, model, Year, Age, kmDriven, Transmission, Owner, FuelType, PostedDate, AdditionInfo, and AskPrice.

Field	Role
Brand	Vehicle manufacturer / brand
model	Vehicle model
Year	Manufacturing/model year
Age	Vehicle age in years
kmDriven	Distance driven
Transmission	Manual or Automatic
Owner	Ownership category
FuelType	Diesel, Petrol, or Hybrid/CNG
PostedDate	Listing month and year
AdditionInfo	Additional listing information
AskPrice	Asking price in Indian Rupees

Dataset source: Kaggle — Used Car Dataset by Mohit Kumar (mohitkumar282).

3. Data Quality Checks & Cleaning

The dataset was checked for duplicate rows, missing/invalid numerical values, unsuitable data types, and suspicious ranges before visualization.

Check / action	Result / decision
Original size	9,582 rows × 11 columns
Duplicate rows	724 identified and removed
After duplicate removal	8,858 rows
kmDriven	Removed commas and ' km'; converted to numeric
Invalid/missing kmDriven after conversion	46 records removed
AskPrice	Removed ₹ and commas; converted to numeric
PostedDate	Converted from month-year text to monthly periods
Negative Age	0 records
Negative kmDriven	0 records
Zero kmDriven	20 records; retained as potentially suspicious
Zero/negative AskPrice	0 records
Future years beyond 2024	0 records
Final cleaned size	8,812 rows × 11 columns

Cleaning decisions

• Duplicates were removed to avoid repeated listings influencing the analysis.

• Mileage and price were converted from formatted text into numeric values.

• Rows with missing mileage after conversion were removed because mileage is a core variable in the EDA.

• Potentially unusual observations were inspected rather than automatically deleted; zero-kilometre records were retained.

• Year + Age was consistently equal to 2024, supporting internal consistency of the age field.

4. Descriptive Analysis

Numerical summary of the cleaned dataset

Statistic	Year	Age	kmDriven	AskPrice (₹)
Count	8,812	8,812	8,812	8,812
Mean	2016.39	7.61	71,034	1,042,024
Median	2017	7	65,137.5	590,000
Std. deviation	4.12	4.12	57,033	1,653,017
Minimum	1986	0	0	15,000
25%	2014	5	43,000	350,000
75%	2019	10	86,000	1,075,000
Maximum	2024	38	980,002	42,500,000

Categorical distributions

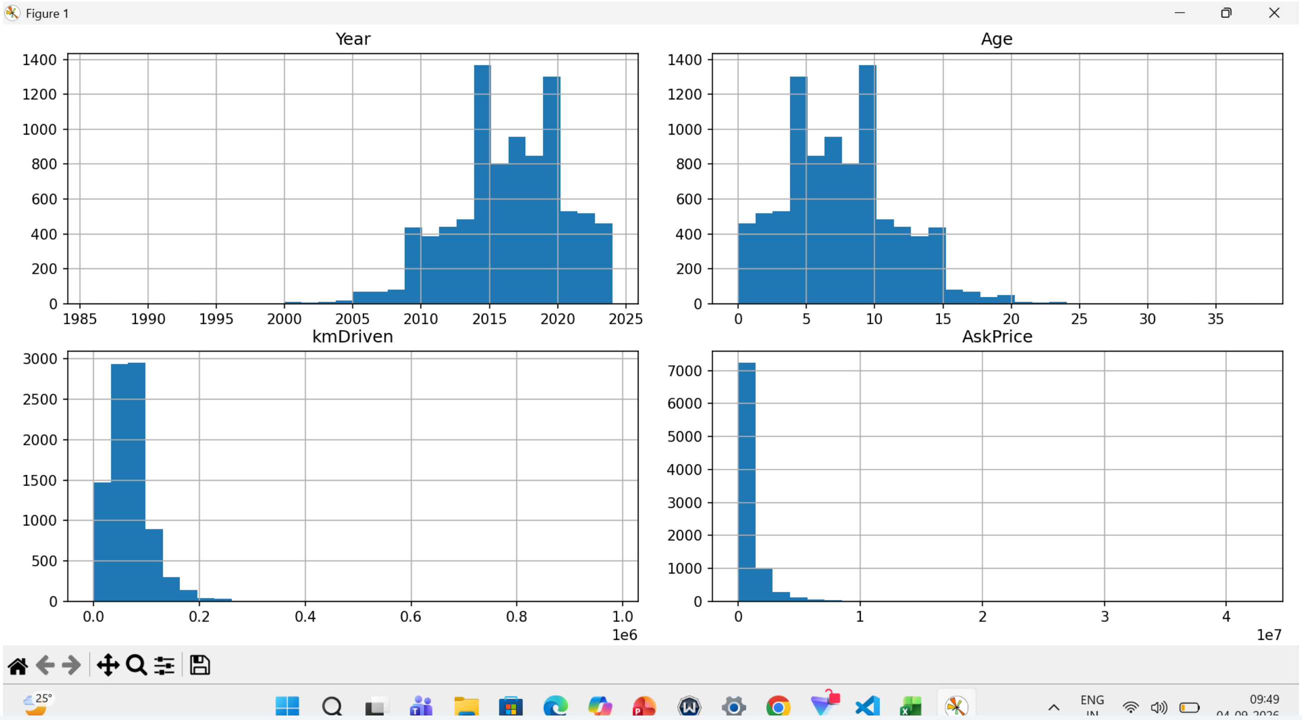
Category	Observed distribution
Top brands	Maruti Suzuki 2,570; Hyundai 1,431; Honda 727; Toyota 719; Mahindra 542; Tata 367
Transmission	Manual 4,640; Automatic 4,172
Fuel type	Diesel 3,508; Petrol 3,497; Hybrid/CNG 1,807
Owner	First owner 4,579; Second owner 4,233

Brand-level asking price summary

Brand	Listings	Mean ₹	Median ₹
Aston Martin	1	26,400,000	26,400,000
Rolls-Royce	2	21,367,500	21,367,500
Bentley	1	18,500,000	18,500,000
Maserati	1	8,000,000	8,000,000
Porsche	18	7,010,556	7,450,000
Lexus	19	6,882,895	5,600,000
Land Rover	70	5,541,700	3,874,999.5
Mercedes-Benz	347	3,566,829	2,450,000
BMW	256	3,037,444	2,197,500
Mini	33	2,962,394	2,850,000

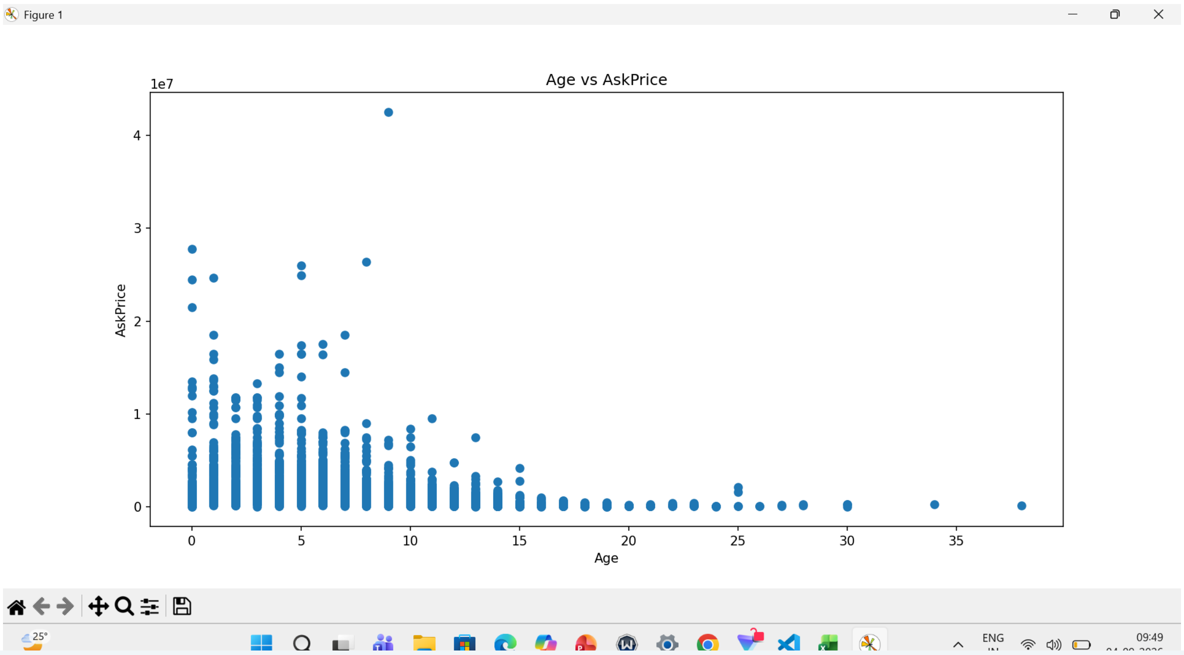
5. Visual Exploratory Data Analysis

Chart 1 — Numerical distributions



Observation: Model years are concentrated in recent years, while vehicle age is concentrated around roughly 4–10 years. Mileage and asking price are strongly right-skewed, with most observations at lower values and a smaller number of high-value observations.

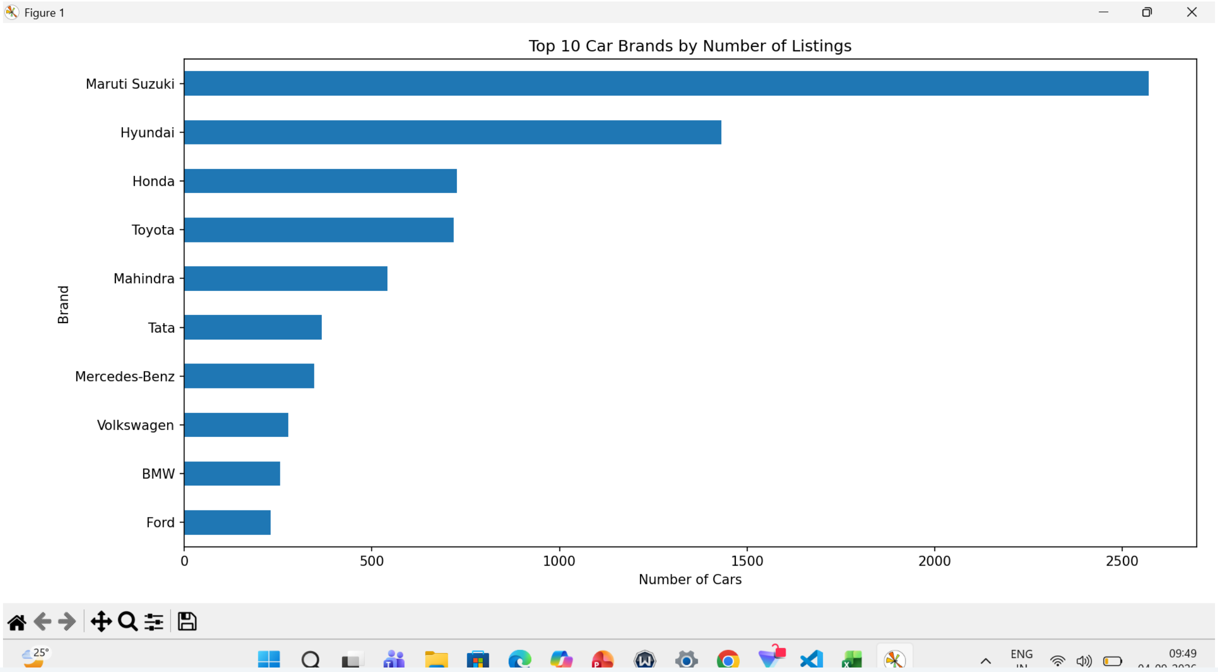
Chart 2 — Age vs AskPrice



Observation: Asking price generally decreases as vehicle age increases. Newer cars show a wider price range, while older vehicles are concentrated at lower prices. High-price observations appear as outliers.

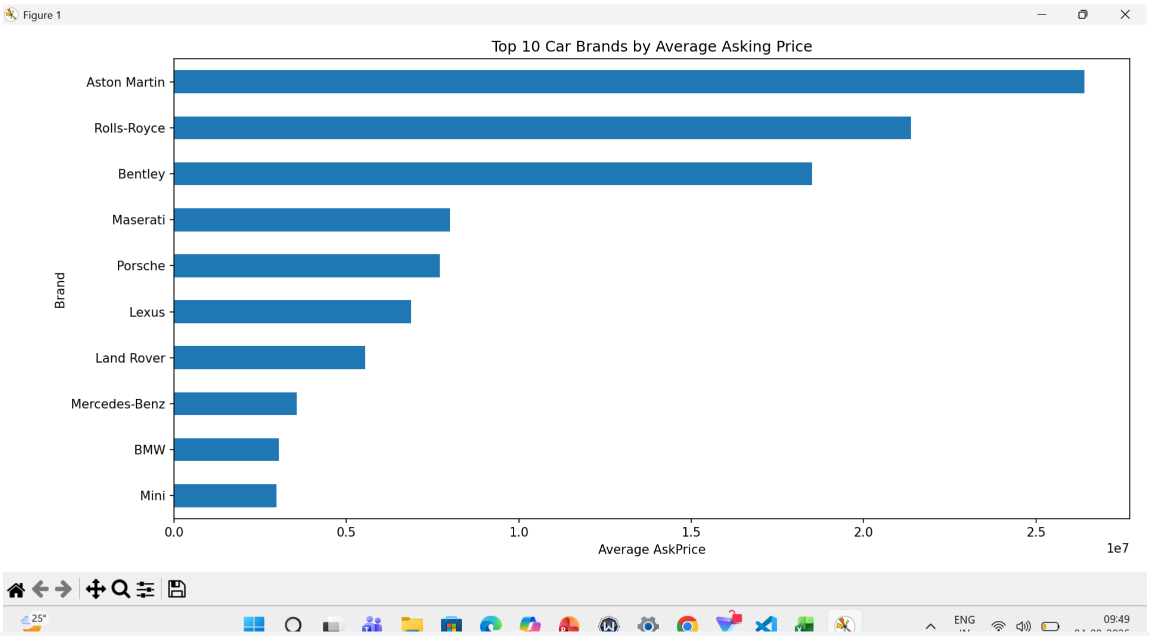
5. Visual Exploratory Data Analysis (continued)

Chart 3 — Top 10 brands by number of listings



Observation: Maruti Suzuki has the largest number of listings (2,570), followed by Hyundai (1,431). Honda and Toyota are the next largest groups, so the dataset is strongly represented by mainstream brands.

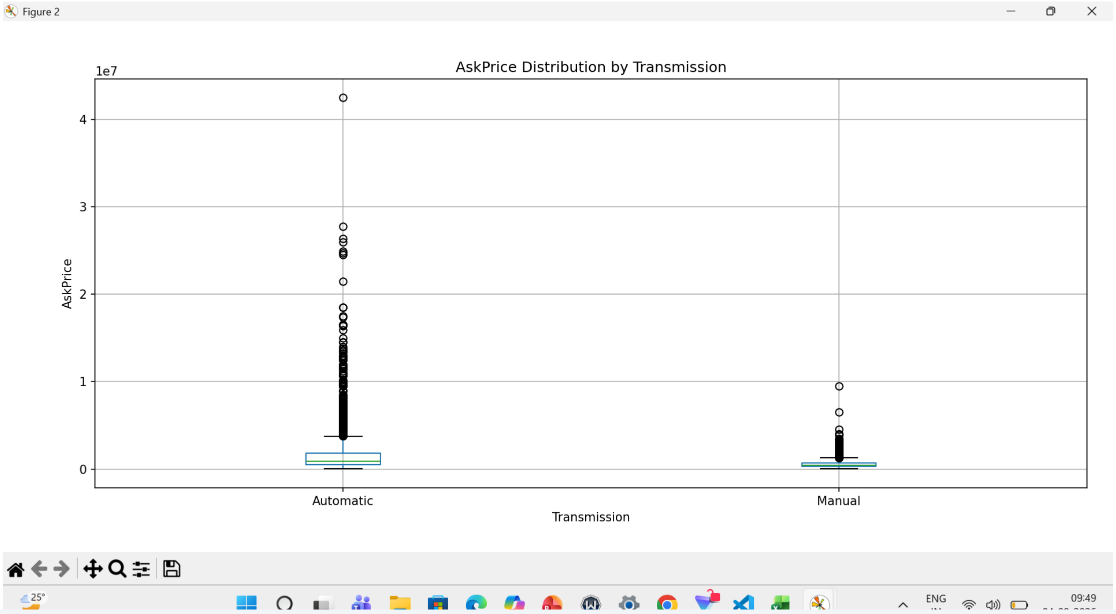
Chart 4 — Top 10 brands by average asking price



Observation: Luxury brands have the highest average asking prices. These estimates should be interpreted cautiously because some brands have only one or two listings.

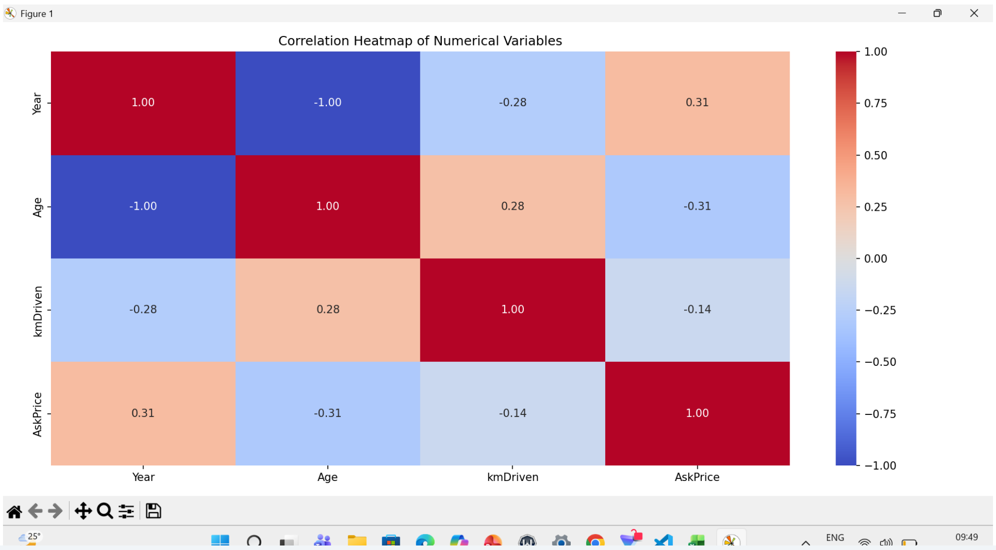
5. Visual Exploratory Data Analysis (continued)

Chart 5 — AskPrice distribution by transmission



Observation: Automatic cars have a higher central asking-price level than manual cars and contain more extreme high-price observations. Both groups contain substantial outliers.

Chart 6 — Correlation heatmap



Observation: Year and Age have a -1.00 correlation because Age is derived from Year. AskPrice has a +0.31 correlation with Year, -0.31 with Age, and -0.14 with kmDriven.

6. Key Findings

• Listing concentration: Maruti Suzuki is the most represented brand with 2,570 listings, followed by Hyundai with 1,431.

• Price skew: The median asking price is ₹590,000 versus a mean of about ₹1.04 million, showing that expensive listings pull the mean upward.

• Age-price relationship: Age has a -0.31 correlation with asking price, and the scatter plot shows generally lower prices for older vehicles.

• Model year: Year has a +0.31 correlation with asking price, consistent with newer vehicles generally being listed at higher prices.

• Transmission: Automatic listings have a higher mean asking price (about ₹1.58 million) than manual listings (about ₹558,227).

• Mileage: kmDriven has only a weak negative correlation with asking price (-0.14), suggesting mileage alone does not explain most price variation.

7. Limitations

• The dataset contains asking prices, not confirmed final transaction prices.

• Luxury-brand averages can be unstable when based on very few listings.

• Correlation indicates association, not causation.

• Factors such as exact variant, condition, location, service history, and accident history may not be fully captured.

8. Conclusion

The exploratory analysis cleaned and examined 8,812 used-car records. The results reveal strong representation of mainstream brands, right-skewed asking prices and mileage, and meaningful relationships between vehicle age, model year, transmission, and asking price. The analysis demonstrates how descriptive statistics and multiple visualization types can reveal useful patterns in real-world automobile data.

Dataset source: Kaggle — Used Car Dataset by Mohit Kumar (mohitkumar282).